9. The floor of the circular swimming pool whose radius is 7 m has to be cemented at the rate of ₹18 per m^2. Find the total cost of cementing the floor.

Objective type questions

10. The formula used to find the area of the circle is _____ *sq.units*.

(i) $4\pi r^2$ (ii) πr^2 (iii) $2\pi r^2$ (iv) $\pi r^2 + 2r$

11. The ratio of the area of a circle to the area of its semicircle is

(i) 2:1 (ii) 1:2 (iii) 4:1 (iv) 1:4

12. Area of a circle of radius 'n' units is

(i) $2\pi r^p$ *sq. units* (ii) πm^2 *sq. units* (iii) πr^2 *sq. units* (iv) πn^2 *sq. units*

2.5 Area of Pathways

We come across pathways in different shapes. Here we restrict ourselves to two kinds of pathways namely circular and rectangular.

2.5.1. Circular Pathways

We observe around us circular shapes where we need to find the area of the pathway. The area of pathway is the difference between the area of outer circle and inner circle. Let 'R' be the radius of the outer circle and 'r' be the radius of inner circle.



Fig. 2.22

Therefore, the area of the circular pathway $= \pi R^2 - \pi r^2$

$= \pi(R^2 - r^2)$ *sq. units.*

2.5.2 Rectangular Pathways

Consider a rectangular park as shown in Fig 2.23. A uniform path is to be laid outside the park. How do we find the area of the path? The uniform path including the park is also a rectangle. If we consider the path as outer rectangle, then the park will be the inner rectangle. Let l and b be the length and breadth of the park. Area of the park (inner rectangle) $= l\ b$ *sq.units*. Let w be the width of the path. If L, B are the length and breadth of the outer rectangle, then $L = l + 2w$ and $B = b + 2w$.



Fig. 2.23

Here, the area of the rectangular pathway

= Area of the outer rectangle – Area of the inner rectangle

= $(LB - lb)$ *sq. units*

7th_Maths_T2_EM_Chp2.indd 34 09-07-2019 12:19:45